

A Cost-Effective Predictive HR Analytics Framework for Sri Lankan SMEs Using Cloud-Native Serverless AI

Interim Progress Report

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Student Name: Theekshana Gimhan

Student ID: 15002

Degree Programme: BSc (Hons) Management Information Systems

Supervisor: Ms. Thanuja Irugalbandara

University: Kaatsu International University

Faculty: Faculty of Computer Science and Engineering

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Abstract

Small and Medium Enterprises (SMEs) constitute the backbone of the Sri Lankan economy, yet enterprise predictive HR analytics platforms remain financially inaccessible to them, typically costing LKR 240,000–450,000 per year for a 50-employee firm. This project designs, builds, and evaluates a cost-effective, cloud-native predictive HR analytics framework that forecasts employee attrition risk for under LKR 10,000 per month, integrated directly into an existing production HR platform (NexusHR). This interim report documents the progress achieved across the proposal's objectives. A production HR platform has been built and deployed on Google Cloud Run; a complete, reproducible machine-learning pipeline has been implemented using a multi-source weighted training strategy (real Saudi and Russian attrition data plus calibrated synthetic Sri Lankan data) validated against held-out real Sri Lankan survey responses; and two attrition models have been trained, deployed behind an IAM-locked serverless inference endpoint, and integrated into the HR product over an API boundary. The central empirical finding to date is a contrast between a weak cross-cultural transfer model (ROC-AUC 0.64) and a strong locally-trained model on eight psychometric constructs (ROC-AUC 0.94), which directly quantifies the need for Sri-Lanka-specific HR data — the very research gap that motivated the study. A live, in-application weekly Pulse Check now supplies those constructs automatically. Measured idle cost is approximately LKR 140 per month, comfortably within the budget target. The report concludes with an honest account of the deviations from the original proposal, the limitations of the present results, and the updated work plan toward the final dissertation.

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1. Introduction

1.1 Background and Motivation

Small and Medium Enterprises (SMEs) form the backbone of Sri Lanka's economy, contributing approximately 52% of national Gross Domestic Product and employing around 45% of the workforce [1]. Despite this significance, most Sri Lankan SMEs still manage human resources through manual, paper-based, or spreadsheet-driven processes [2]. Under such reactive models, critical decisions — notably identifying employees at risk of resignation — are made only after the loss has occurred, incurring recruitment, onboarding, and knowledge-loss costs.

The global HR-technology market addresses this with AI-powered platforms such as SAP SuccessFactors, Workday, and Oracle HCM Cloud. These typically use per-employee-per-month pricing of roughly USD 8–15, equating to LKR 240,000–450,000 annually for a 50-employee SME [4] — financially out of reach for the firms that would benefit most. Recent advances in cloud-native, serverless computing have changed the economics of deploying machine learning: services that scale to zero enable a pay-per-prediction model that removes idle infrastructure cost [5], [6]. This creates an opportunity to deliver enterprise-grade predictive analytics to resource-constrained SMEs at a fraction of the conventional cost, in alignment with Sri Lanka's National AI and Digital Economy strategies [7].

1.2 Problem Statement

Employee turnover imposes a disproportionate burden on SMEs: replacing a departing employee costs an estimated 50–200% of annual salary once recruitment, onboarding, training, and lost productivity are included [8]. Sri Lanka's intensifying skilled-worker emigration (over 300,000 departures annually) makes retention of remaining staff a survival factor [17]. Although machine learning has been shown to predict attrition effectively [9], [10], adoption among SMEs is low due to three barriers: cost (enterprise PEPM pricing), infrastructure (on-premise AI requires GPUs and DevOps expertise [11]), and integration (standalone tools create data silos disconnected from HR workflows [12]). The research gap is therefore a cost-optimised, cloud-native predictive analytics framework that is (a) financially accessible (under LKR 10,000/month), (b) architecturally integrated into an existing HR system, and (c) contextualised for the Sri Lankan workforce.

1.3 Research Questions

- **RQ1:** What are the most significant predictors of employee attrition derivable for the Sri Lankan SME context, and to what extent do attrition patterns transfer from international data to Sri Lanka?
- **RQ2:** To what extent can a serverless cloud architecture reduce the operational cost of deploying HR predictive analytics relative to persistent infrastructure, and can it remain within a monthly budget of LKR 10,000?
- **RQ3:** Can a hybrid dataset approach — combining real international attrition data with calibrated synthetic data — produce a usable attrition classifier for the low-volume Sri Lankan SME data environment?

1.4 Aim and Objectives

Aim: to design, develop, and evaluate a cost-effective, serverless AI-driven predictive analytics module integrated into an existing cloud-native HR system to forecast employee attrition risk for Sri Lankan SMEs.

- **O1 — Feature identification:** identify behavioural and psychometric predictors of attrition appropriate to the Sri Lankan SME context, in compliance with the Sri Lanka Personal Data Protection Act No. 9 of 2022 [15].
- **O2 — Architecture design:** design a serverless cloud architecture maintaining low monthly operational cost for a 20–50 employee organisation, with a verifiable cost baseline against always-on deployments.
- **O3 — Model development:** develop and train ML classification models using a hybrid dataset strategy, targeting a usable early-warning operating point (high recall on the attrition class).
- **O4 — System integration:** implement an integrated risk-insights interface within the HR product, including SHAP-based explanations.
- **O5 — Evaluation:** evaluate predictive performance, deployment cost, and usability (System Usability Scale).

1.5 Scope

In scope: binary attrition-risk classification; a serverless prediction service; an in-product risk-insights interface with SHAP explanations; cost benchmarking; a calibrated synthetic-data pipeline; and privacy-preserving data handling. Out of scope: real-time communications monitoring, mobile development, changes to core payroll/leave/attendance logic, multi-language NLP, and succession planning.

2. Literature Review (Status)

The literature review established in the proposal has been consolidated and extended during implementation. It is organised into three themes and is substantially complete; remaining work is the integration of explainability and resampling literature surfaced during model development.

Machine Learning for Employee Attrition Prediction

Ensemble tree methods — particularly Random Forest and Gradient Boosting — consistently outperform logistic regression for turnover classification, exceeding 85% accuracy on benchmark data [9], [10]. Tree ensembles are well suited to tabular HR data because they capture non-linear feature interactions [9]. The IBM HR Analytics dataset (1,470 records, 35 features) is the field's common benchmark [14]. A recurring limitation is reliance on Western corporate datasets with little validation in developing-economy SME settings — a gap this project targets directly.

Serverless and Cloud-Native AI Economics

Comparative studies show serverless architectures offer strong cost and scalability advantages for intermittent workloads [6] — precisely the usage profile of scheduled or on-demand HR predictions. Managed ML platforms enable pay-per-prediction pricing that removes idle compute cost [5], and the Machine-Learning-as-a-Service paradigm lowers the infrastructure-expertise barrier for non-technical organisations [11]. No prior study, however, quantifies the cost of predictive HR analytics under an explicit SME budget constraint (under LKR 10,000/month).

The Digital Divide and HR Analytics Integration in Sri Lankan SMEs

Sri Lankan SMEs face well-documented barriers to digital transformation — limited finance, scarce skills, and weak infrastructure [1], [2], [3] — compounded by skilled-worker emigration that makes retention strategic [17]. Even where analytics are adopted, disconnection from operational HR systems creates silos that undermine practical value [12]. Two methodological pillars adopted during implementation are also grounded in the literature: SHAP for exact, model-agnostic feature attribution [18], and SMOTE-family resampling for class imbalance [19]. Finally, the distinction between turnover intention and observed turnover (the intention–behaviour gap [20]) is central to interpreting the validation data used in this study.

3. Methodology

3.1 Research Approach (Design Science)

The project follows the Design Science Research (DSR) methodology [13], appropriate for development-oriented research in which the construction and evaluation of an IT artifact is the primary contribution. The artifact here is the integrated, cost-effective predictive HR analytics framework. The work proceeds through problem identification, data preparation, model development, architecture design and cloud integration, product integration, and evaluation.

3.2 Data Strategy

No public, individual-level HR attrition dataset exists for Sri Lanka. The closest available proxy is a 2023 survey of 230 Sri Lankan startup employees measuring turnover intention on an ordinal scale [21] — not observed resignations. The model therefore cannot be trained exclusively on Sri Lankan data. The adopted strategy is multi-source weighted training: train on real international data from developing economies with comparable workforce dynamics, augment with calibrated synthetic data carrying Sri Lankan feature distributions, and validate exclusively on the held-out real Sri Lankan responses.

Source	Records	Type	Weight	Role
Saudi Employee Attrition	1,191	Real, developing-country private sector	2.0	Training
Russian Employee Turnover	1,129	Real, private-sector company	2.0	Training
Calibrated synthetic (SL context)	500	Simulated, Sri Lankan SME	0.5	Training
Sri Lanka startup survey [21]	230	Real, turnover intention	—	Held-out validation
IBM HR Analytics	1,470	Synthetic benchmark	—	Published comparison only

Table 1: Multi-source data strategy (master training set = 2,820 records).

Three normalisation decisions ensure cross-source comparability. First, monthly income is z-score normalised within each source before merging, so the model learns relative salary position rather than incompatible nominal currencies. Second, ordinal survey features measured on a 1–3 scale in the training data but a 1–5 scale in the Sri Lankan survey are min-max rescaled within each dataset. Third, features not collected by a given source are retained as missing (NaN) rather than zero, preserving the distinction between 'measured as zero' and 'not measured'; median imputation is then applied explicitly at train time. The synthetic data is not hand-coded: a logistic regression is fitted on 2,550 real records to obtain coefficient directions and magnitudes (job satisfaction emerges as the dominant driver), while the intercept is recomputed to target a 15% base attrition rate consistent with Sri Lankan labour-market reports rather than the ~43% international base rate.

3.3 Machine-Learning Approach

- **Model:** scikit-learn Random Forest with SMOTE-family resampling for class imbalance, chosen over managed AutoML for transparency, exact explainability, and near-zero cost.
- **Explainability:** SHAP TreeExplainer (exact) for per-prediction feature attribution [18].
- **Validation:** the 230 Sri Lankan records are kept entirely held out; the local model additionally uses repeated stratified 5-fold cross-validation within that data, reported as a multi-seed mean.
- **Primary metric:** recall on the attrition class (minimising false negatives), reported alongside precision, F1, ROC-AUC, PR-AUC, and confusion matrices — never recall in isolation.

4. Design and Implementation Progress

4.1 System Architecture

The framework uses a two-system design connected over an API boundary rather than a monolith. The first system is the production HR platform (React 19 front end; Node.js / Express 5 / Prisma 6 back end; PostgreSQL), deployed on Google Cloud Run. The second is a standalone machine-learning inference microservice (Python / FastAPI) that serves the trained models. The HR back end calls the inference service over authenticated HTTP. This separation keeps the cost-sensitive ML component independently deployable and allows both systems to scale to zero when idle.

4.2 The HR Platform

The base HR platform is production-grade and operational, providing payroll (with Sri Lankan EPF, ETF, and PAYE calculation), leave, attendance, and document management, with multi-tenant isolation and role-based access control. It is continuously deployed to Cloud Run via an automated pipeline. This platform supplies both the operational context for the analytics module and the user interface in which risk insights are surfaced.

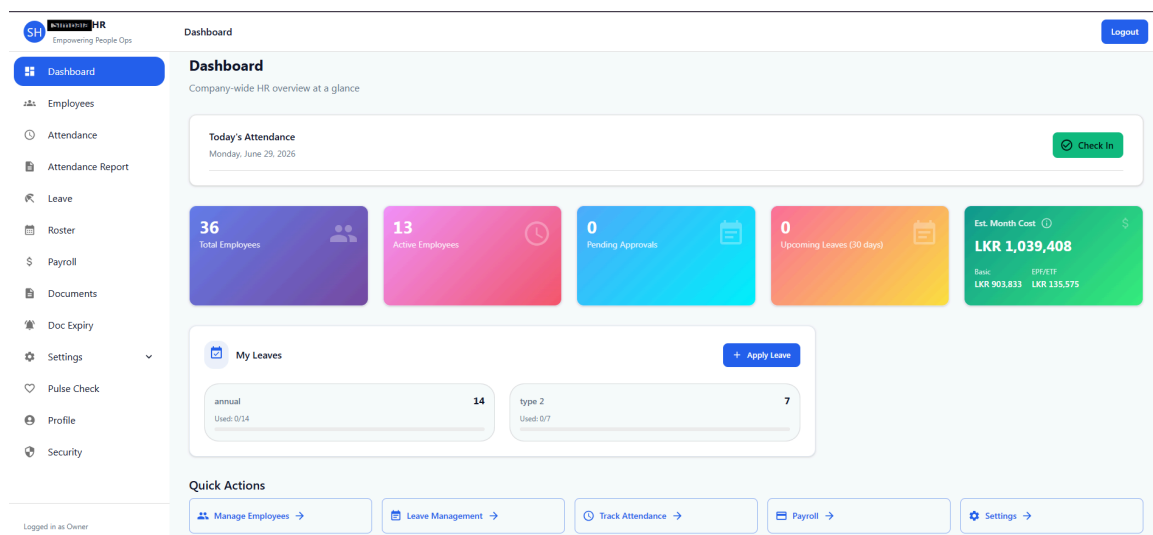


Figure 1: The NexusHR / Simpala HR management dashboard, deployed on Google Cloud Run.

4.3 Machine-Learning Pipeline

A complete, reproducible five-stage pipeline has been implemented: dataset acquisition; raw preprocessing (including robust multi-encoding handling for the Cyrillic Russian source); calibration of the logistic coefficients used for synthetic generation; calibrated synthetic-data generation; and a merge-and-clean stage that assembles the 2,820-record master training set, the 230-record held-out Sri Lankan validation set, and the IBM benchmark. The merge stage asserts target integrity (binary, non-missing labels, with every source contributing labelled rows) before writing. A training stage then fits the models, performs cross-validation, computes SHAP attributions, and emits a structured training report.

4.4 Serverless Inference and HR-App Integration

Both models are deployed behind an IAM-locked Cloud Run inference service that scales to zero. The service exposes health, model-info, and prediction routes, returning for each request the probability, decision threshold, risk band, and per-request SHAP contributions. Train/serve parity is enforced by mirroring the training-time preprocessing (encoding, persisted rescale bounds, and median imputation) exactly in the serving code. The HR back end integrates with this service through a dependency-free proxy that mints a Google-signed identity token from the Cloud Run metadata server and forwards requests; the corresponding risk-insights card is surfaced on the employee detail page of the HR front end, gated by a dedicated permission. A cold-start tolerance (extended timeout plus a single retry) was added so that the first request after idle does not fail, while preserving scale-to-zero.

Attrition risk
Predictive model (beta)

Inputs are employee-sentiment constructs (1-5). Their live source is the planned Pulse Check survey; for now enter or sample them below. The model estimates turnover **intention**, not a certainty — use it as an early-warning signal.

Job satisfaction	3
Work-life balance	3
Happiness	3
Management support	3
Career management	3
Innovative work behaviour	3
Leader-member exchange	3
Coworker support	3

[Predict risk](#) [Reset](#)

Logged in as Owner

Figure 2: The Attrition Risk card on the employee detail page (manual what-if mode): an authorised manager enters the eight sentiment constructs (1-5) and requests an on-demand prediction. Its automated counterpart, driven by the weekly Pulse Check, is shown in Figure 4.

4.5 The Pulse Check — Automated Construct Capture

The strongest model consumes eight psychometric constructs whose natural production source is an employee survey rather than operational HR records. To supply these automatically and at low cost, a lightweight in-application weekly Pulse Check has been implemented and deployed: a 16-item Likert micro-survey (two items per construct) whose responses are averaged into the eight constructs using the same mean-of-items definition used in training, then scored against the inference service and stored once per employee per week. This was chosen in preference to a heavier conversational-AI agent specifically to preserve the cost objective. By design, employees see only a confirmation and never their own risk score; the resulting risk signal is visible to managers. Scoring is best-effort, so a model outage never blocks a submission.

Weekly Pulse Check
A 2-minute check-in on how work feels this week

✓ You've completed this week's Pulse Check. Thank you! You can update your answers below if anything has changed.

ⓘ Your honest answers help us understand and improve the employee experience. Tell us how much you agree with each statement (1 = strongly disagree, 5 = strongly agree).

1. I feel satisfied with my current job.

Disagree Neutral Agree

2. My work gives me a sense of accomplishment.

Figure 3: The employee weekly Pulse Check: a 16-item Likert micro-survey that feeds the eight model constructs automatically.



Figure 4: Manager view: the latest Pulse-derived attrition risk for an employee, on the employee detail page.

5. Preliminary Results and Evaluation

5.1 Headline Result: Transfer vs Local

The training stage reports two models side by side; the contrast between them is the project's central empirical contribution to date.

Model	Features	SL ROC-AUC	SL PR-AUC	Usable operating point
Transfer (international → SL)	4 shared	0.64	0.29	None — recall 0.80 forces precision ~0.15
Local (within SL, 8 constructs)	8	0.94	0.79	Precision 0.73 / Recall 0.82 at threshold 0.30

Table 2: Cross-cultural transfer model vs locally-trained model on the held-out Sri Lankan data.

Cross-cultural transfer of attrition patterns is weak (ROC-AUC 0.64, barely above chance), whereas a locally-trained model on rich, same-instrument psychometric features is strong and usable (ROC-AUC 0.94, with a genuine early-warning operating point catching 27 of 33 at-risk respondents). This directly quantifies why Sri Lanka requires its own HR data — the precise gap that motivated the project. The weak transfer result is not a failure; it is the evidence. The local model's most influential constructs (work-life balance, co-worker support, innovative work behaviour, and job satisfaction) are interpretable and consistent with the turnover literature.

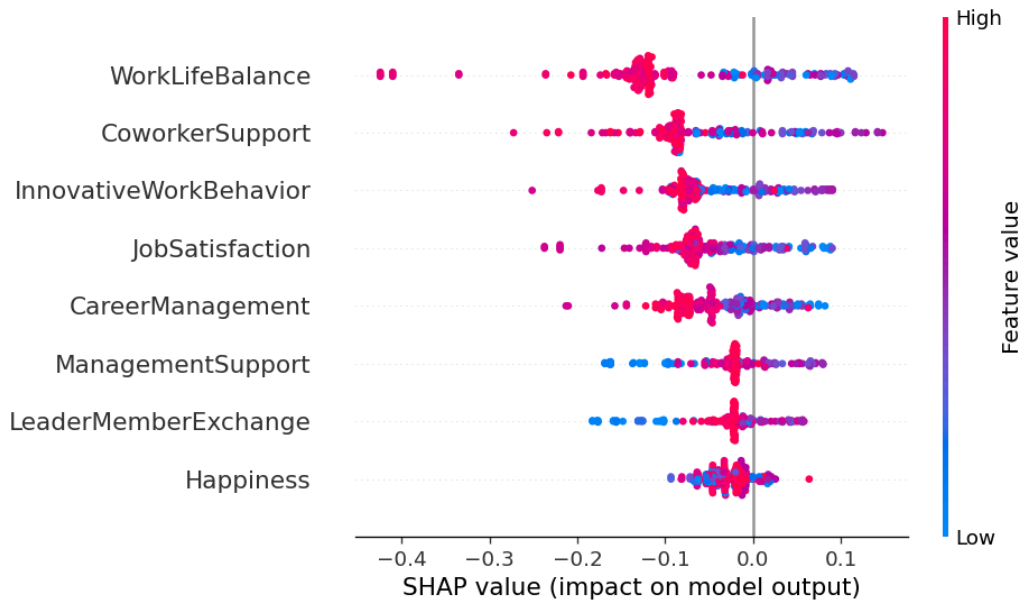


Figure 5: SHAP summary (beeswarm) for the local model. Each point is one respondent; position shows the construct's contribution to predicted flight-risk intention, colour shows the construct value (blue = low, red = high).

Lower work-life balance, co-worker support, and job satisfaction push risk upward, as expected.

5.2 Cost Evaluation

Because both the HR platform and the ML inference service run on Cloud Run and scale to zero, idle cost is effectively nil. The measured/estimated operational cost is approximately LKR 140 per month for a representative 50-employee workload (daily attendance logging, weekly pulse checks, ~100 predictions/month, and one retraining run). Even at ten times this usage the cost remains under LKR 1,500 — well within the LKR 10,000 target, providing strong preliminary evidence for RQ2. A formal three-architecture comparison using exported billing data is scheduled for the evaluation phase.

5.3 Honest Limitations

The following are properties of the available data, not defects, and they bound the present claims:

- **Small N for the local model:** 230 records with ~33 positives; the 0.94 ROC-AUC is reported as a repeated cross-validation mean (range 0.93–0.94) and is encouraging rather than definitive.
- **Intention, not behaviour:** both models predict turnover intention (composite ≥ 3.5), not observed resignations; results are a flight-risk proxy subject to the intention–behaviour gap [20].
- **Differing feature provenance:** the transfer model is limited to the four internationally-shared features (dominated by age), while the local model's eight constructs are survey-sourced; the two target different deployment paths.
- **Base-rate mismatch:** international training prevalence (~42%) differs from the Sri Lankan rate (14.3%); thresholds are tuned on a held-out split and the full metric suite is reported at each operating point.

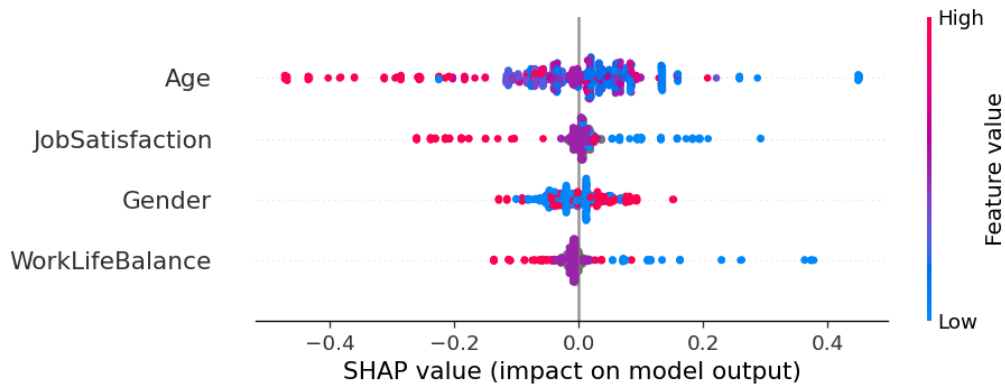


Figure 6: SHAP summary for the cross-cultural transfer model. With only the four internationally-shared features, the model leans heavily on Age, illustrating why transfer alone is a weak basis for Sri Lankan attrition prediction.

Accordingly, the thesis will lead with the transfer-vs-local comparison as the empirical contribution rather than a single universal recall headline. Obtaining real Sri Lankan attrition (not intention) data from a partner SME is identified as the highest-value future work.

6. Progress Summary, Deviations and Updated Plan

6.1 Progress Against Objectives

Objective	Status	Evidence
O1 Feature identification	Largely complete	Calibrated predictors + 8 validated psychometric constructs identified; PDPA-compliant handling.
O2 Architecture design	Complete (preliminary cost shown)	Serverless two-system architecture deployed; idle cost ~LKR 140/month.
O3 Model development	Complete (first iteration)	Two models trained; local model usable at P0.73/R0.82; SHAP integrated.
O4 System integration	Complete	Inference service + HR-app proxy + risk card + live Pulse Check.
O5 Evaluation	In progress	Predictive metrics done; formal cost comparison + SUS study pending.

Table 3: Progress against the proposal's objectives.

6.2 Deviations from the Proposal and their Justification

Implementation surfaced evidence that justified several deliberate, defensible deviations from the original plan. These are reported transparently as they form part of the research narrative.

- **Inference platform:** Cloud Run with a self-hosted scikit-learn model replaced Vertex AI AutoML. AutoML is costly per training run, opaque, and incompatible with exact SHAP; a transparent Random Forest gives comparable accuracy at near-zero cost, better serving both the cost objective and the explainability requirement.
- **Training data:** real multi-source international data (Saudi + Russian) replaced reliance on the IBM dataset for training. IBM data is itself synthetic with fictional income units; it is retained as a published benchmark only, while real attrition events drive training.
- **Predictive features:** the strongest signal came not from operational HR data alone but from psychometric constructs. This motivated the two-model design and the Pulse Check as the construct source — a finding that strengthens, rather than weakens, the contribution.
- **Framing of the headline metric:** rather than asserting a single '>80% recall' result, the project reports the transfer-vs-local contrast, which is a more honest and more original contribution given the data reality.

6.3 Risk Register

Risk	Likelihood	Impact	Mitigation
No real SL attrition (vs intention) data	High	Medium	Framework/methodology is the contribution; pipeline accepts partner data at highest weight without redesign.
Small validation N limits claims	High	Medium	Repeated CV with seed range; report full metric suite and limitations explicitly.
GCP cost exceeds target	Low	Medium	Scale-to-zero services; billing alerts; large safety margin already demonstrated.
SUS below target	Medium	Medium	Pilot with 2–3 users and iterate dashboard UX before the formal study.
PDPA compliance gap	Low	High	PII stripped at the database layer; no identifiers in features; audited exports.

Table 4: Principal project risks and mitigations.

6.4 Updated Work Plan

Period	Planned activity
Jul 2026	Formal three-architecture cost comparison from exported GCP billing data; finalise evaluation tables and figures.
Jul 2026	Usability evaluation (System Usability Scale) with 5–10 SME stakeholder personas; incorporate feedback.
Jul–Aug 2026	Optional data-protection hardening (automated PII scanning) and analytics queries; pursue a partner SME for real attrition data if feasible.
Aug 2026	Write and format the final dissertation; plagiarism check; supervisor review at least one week before submission (final report due 31 August 2026).
Sep 2026	Final presentation and demonstration.

Table 5: Updated work plan to final submission.

In summary, the project has reached a mature state well beyond a typical interim stage: a deployed product, a reproducible ML pipeline, a live serverless inference service with in-product integration, and a clear, honest empirical finding. The remaining work is principally formal evaluation (cost and usability) and the writing of the final dissertation.

References

- [1] Department of Census and Statistics, "Sri Lanka Labour Force Survey — Annual Report 2023," Ministry of Finance, Sri Lanka, 2023.
- [2] S. Kirupananthan, M. Kayathiry, and S. Niroshan, "Exploring Challenges for Artificial Intelligence Readiness in SMEs in Sri Lanka," *Vavuniya Journal of Business Management*, vol. 4, no. 1, 2024.
- [3] Information and Communication Technology Agency of Sri Lanka (ICTA), "Sri Lanka Digital Economy Report 2023," ICTA, Colombo, 2023.
- [4] B. Charney, "HR Software Pricing Guide: How Much Does HR Software Cost?," *People Managing People*, 2024.
- [5] Google Cloud, "Vertex AI: Train and Deploy ML Models," Google LLC, 2024.
- [6] N. Kodakandla, "Serverless Architectures: A Comparative Study of Performance, Scalability, and Cost in Cloud-native Applications," *Iconic Research and Engineering Journals*, vol. 5, no. 2, pp. 136–150, 2021.
- [7] Ministry of Technology, "Artificial Intelligence in Sri Lanka: White Paper," Government of Sri Lanka, 2024.
- [8] D. G. Allen, "Retaining Talent: A Guide to Analyzing and Managing Employee Turnover," SHRM, Alexandria, VA, 2008.
- [9] I. H. Sarker, "Machine Learning: Algorithms, Real-World Applications and Research Directions," *SN Computer Science*, vol. 2, no. 3, pp. 1–21, 2021.
- [10] R. Punnoose and P. Ajit, "Prediction of Employee Turnover in Organizations using Machine Learning Algorithms," *Int. J. Advanced Research in Artificial Intelligence*, vol. 5, no. 9, pp. 22–26, 2016.
- [11] M. Ribeiro, K. Grolinger, and M. A. M. Capretz, "MLaaS: Machine Learning as a Service," in *Proc. IEEE ICMLA*, 2015, pp. 896–902.
- [12] D. Angrave, A. Charlwood, I. Kirkpatrick, M. Lawrence, and M. Stuart, "HR and Analytics: Why HR is Set to Fail the Big Data Challenge," *Human Resource Management Journal*, vol. 26, no. 1, pp. 1–11, 2016.
- [13] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design Science in Information Systems Research," *MIS Quarterly*, vol. 28, no. 1, pp. 75–105, 2004.
- [14] IBM, "IBM HR Analytics Employee Attrition & Performance Dataset," Kaggle, 2017.
- [15] Parliament of Sri Lanka, "Personal Data Protection Act, No. 9 of 2022," Government Gazette of the Democratic Socialist Republic of Sri Lanka, 2022.
- [16] J. Brooke, "SUS — A Quick and Dirty Usability Scale," in *Usability Evaluation in Industry*, London: Taylor & Francis, 1996, pp. 189–194.
- [17] Sri Lanka Bureau of Foreign Employment, "Annual Statistical Report of Foreign Employment 2023," SLBFE, Colombo, 2023.

- [18] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 4765–4774.
- [19] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [20] R. W. Griffeth, P. W. Hom, and S. Gaertner, "A Meta-Analysis of Antecedents and Correlates of Employee Turnover: Update, Moderator Tests, and Research Implications for the Next Millennium," *Journal of Management*, vol. 26, no. 3, pp. 463–488, 2000.
- [21] L. Kanchana and R. Jayathilaka, "Factors impacting employee turnover intentions among professionals in Sri Lankan startups," *PLOS ONE*, vol. 18, no. 2, e0281729, 2023, doi: 10.1371/journal.pone.0281729.